

AI – Powered Employee Attrition Prediction and Workforce Intelligence System

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Index

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What is Employee Attrition

Employee attrition means employees leaving an organization or company.

Employees may leave for different reasons, such as:

- Low salary or better job opportunities
- Lack of job satisfaction
- High workload or overtime
- Poor work-life balance
- Lack of career growth
- Problems with management or workplace
- Relocation or personal reasons

Research Gap

- **Gap 1:**

Existing research mainly focuses on employee attrition prediction using ML models.

- **Gap 2:**

Explainable AI (SHAP) is used to identify factors influencing predictions.

- **Gap 3:**

Limited focus on converting predictions into actionable retention strategies.

- **Gap 4:**

Limited integration of prediction, explanation, recommendations, and managerial decision support into one system.

Problem Statement

"Organizations need a proactive way to identify and manage employee attrition before valuable employees leave."

- Employee attrition leads to higher recruitment and replacement costs.
- Employee turnover can cause productivity and workforce disruption.
- HR may have difficulty identifying potential leavers early.
- Existing approaches may provide predictions but limited support for retention decisions.

Methodology

1. Employee Data Management

Accepts and validates employee information entered by HR.

2. Attrition Prediction

Processes the employee data using the trained ANN model to predict whether the employee is likely to leave and generate a risk score.

3. Explainable AI

Uses SHAP to identify and display the key factors influencing the attrition prediction.

4. Smart Recommendations

Uses the identified risk factors to suggest suitable employee retention strategies for HR.

5. Manager Decision Summary

Combines the prediction, risk factors, and recommendations to generate a concise AI-based summary for managers.

6. HR Dashboard

Visualizes employee attrition trends, risk distribution, and other workforce analytics to support HR decision-making.

Literature Survey

Sr. No.	Title of the Paper	Author / Publication / Year	Technique Used	Summary
1.	Employee Attrition Prediction Using Bayesian Optimized Stacked Ensemble Learning and Explainable AI	Karabi Konar et al. / SN Computer Science (Springer) / 2025	- Bayesian Optimization - Stacked Ensemble - SHAP	Proposes an optimized ensemble learning model with Explainable AI to improve attrition prediction accuracy and interpretability for HR managers
2.	HR Analytics: Predicting Employee Turnover using Machine Learning Models	IEEE ICDISS Conference / 2025	- Random Forest - Gradient Boosting - Logistic Regression	Compares multiple ML models for employee turnover prediction and demonstrates how predictive analytics supports proactive HR decision-making.
3.	Mitigating Attrition: Data-Driven Approach Using Machine Learning and Data Engineering	Naveen Edapurath Vijayan / arXiv / 2025	- Machine Learning - Feature Engineering - SHAP	Presents a data engineering and machine learning framework for predicting attrition with interpretable insights to support retention strategies.

Literature Survey

Sr. No.	Title of the Paper	Author / Publication / Year	Technique Used	Summary
4.	Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics	Maytha Al-Ali et al. / Scientific Reports (Nature) / 2026	• Adaptive Boosting • Histogram Gradient Boosting • SHAP	Integrates advanced ML with Explainable AI to predict employee attrition and identify the most influential factors affecting employee turnover.
5.	Explainable AI for Employee Turnover Prediction: A SHAPBased Intelligent Analytics Approach	Haojie Jia / Scientific Reports (Nature) / 2026	• SHAP • Explainable AI	Focuses on transparent employee turnover prediction using SHAP while considering fairness, threshold calibration, and responsible AI practices.
6.	Explainable Attrition Risk Scoring for Managerial Retention Decisions in Human Resource Analytics	M. S. Pavithran & S. M. Vadivel / Frontiers in Big Data / 2026	• Explainable AI • Predictive Analytics • Risk Scoring	Develops an explainable attrition risk scoring system to help managers identify at-risk employees and improve retention decisions.

Scope & Objective

- Predict employee attrition using AI and machine learning techniques.
- Identify the key factors influencing employee attrition through Explainable AI.
- Provide smart retention recommendations to help HR reduce employee turnover.
- Generate manager decision summaries for better workforce planning.
- Visualize HR insights using an interactive analytics dashboard.
- Support data-driven HR decision-making to improve employee retention and organizational performance.

Project Contribution in Society

1. Improved Employee Retention

Helps organizations identify attrition risks early and take timely retention actions.

2. Reduced Organizational Costs

Helps reduce recruitment and replacement costs caused by employee turnover.

3. Better Workplace Decisions

Provides HR and managers with data-driven insights for informed decision-making.

4. Healthier Workforce Management

Encourages organizations to understand employee concerns and improve overall workforce well-being.

Future Scope

1. Real-Time HR Data Integration

Integrate live HR data sources for continuously updated employee information.

2. Advanced AI & Larger Datasets

Use larger datasets and advanced AI models to improve prediction accuracy and recommendations.

3. Fair & Responsible AI

Enhance fairness, bias detection, and responsible AI practices for reliable HR decisions.

Conclusion

The proposed AI-Powered Employee Attrition Prediction and Workforce Intelligence System aims to help organizations proactively identify employees at risk of leaving. By combining real-time employee input, ANN-based prediction, Explainable AI using SHAP, smart retention recommendations, and managerial decision support, the system transforms employee data into actionable insights. The integrated platform can help HR teams make timely, data-driven decisions, improve employee retention, and reduce the impact and cost of workforce attrition.

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Thank
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